

Problem 1: Toy Factory – Double the Toy Count

A toy factory is preparing gift boxes for a festival. Every toy produced today will be packed in pairs, so the number of toys from each production batch needs to be doubled.

Write a function `doubleAll(numbers)` that returns a new array where each batch count is doubled.

Examples

```
doubleAll([4, 2, 3]);
```

```
// Returns: [8, 4, 6]
```

```
doubleAll();
```

```
// Returns: []
```

```
doubleAll([1, 5, 2]);
```

```
// Returns: [2, 10, 4]
```

Problem 2: Online Bookstore – Count Book Searches

An online bookstore records every book title searched by customers throughout the day. The same title may be searched multiple times. The store wants a summary showing **how many times each book title was searched**.

Write a function `countWords(input)` that counts the occurrences of each book title in the given array.

Use the following constructor to represent each result.

```
function Counter(word, count) {
```

```
  // code
```

```
}
```

Function Signature

```
function countWords(input) {
```

```
}
```

Parameters

- `input` – An array of book titles (strings).

Returns

An array of Counter objects where:

- `word` represents the book title.
- `count` represents the number of times the title appears.

Example

Input

```
var inputBooks = [  
  "Harry Potter",  
  "The Hobbit",  
  "Harry Potter",  
  "The Alchemist",  
  "The Hobbit",  
  "Harry Potter"  
];
```

Output

```
[  
  { word: "Harry Potter", count: 3 },  
  { word: "The Hobbit", count: 2 },  
  { word: "The Alchemist", count: 1 }  
]
```

Constraints

- The input array may contain duplicate book titles.
- String comparison is case-sensitive.
- If the input array is empty or not provided, return an empty array.

3) Product Code Generator

An inventory system stores product codes in the format **LAPTOP-HP-4589**. The warehouse team needs only the product category (the first word before the first hyphen).

Write a function **getCategory(code)** that returns the product category.

Hint : use substring

Examples

```
getCategory("LAPTOP-HP-4589"); // "LAPTOP"  
getCategory("MOBILE-SAMSUNG-22");// "MOBILE"
```

4) Movie Title Search (indexOf)

A streaming platform allows users to search whether a keyword appears in a movie title.

Write a function **findKeyword(title, keyword)** that returns the starting position of the keyword.

Examples

```
findKeyword("The Dark Knight", "Dark"); // 4
```

```
findKeyword("Avengers Endgame", "Game"); // -1
```

5) Invoice Number Analyzer

An invoice number may contain multiple hyphens. The finance team wants to know where the **last hyphen** appears.

Write a function **findLastSeparator(invoiceNo)**.

Hint : use **(lastIndexOf)**

Examples

```
findLastSeparator("INV-2026-001"); // 8
```

```
findLastSeparator("A-B-C"); // 3
```

6) Username Availability

A website does not allow usernames containing the word **admin**.

Write a function **containsRestrictedWord(username)** that returns whether the restricted word exists.

Hint : use **includes**

Examples

```
containsRestrictedWord("admin123"); // true
```

```
containsRestrictedWord("john123"); // false
```